

Machine Learning Project

Project 2 Report

Table 1: Validation Accuracy of Bagged Decision Trees for Different Values of n_estimators and max_depth

Dataset	10	10	10	20	20	20	50	50	50	100	100	100
max_depth	5	10	None	5	10	None	5	10	None	5	10	None
c300_d100	0.6382	0.6231	0.6231	0.6834	0.6633	0.6633	0.7286	0.7035	0.7035	0.6884	0.6834	0.6834
c300_d1000	0.8074	0.7589	0.7589	0.8289	0.7849	0.7819	0.8519	0.8464	0.8449	0.8534	0.8674	0.8689
c300_d5000	0.7877	0.8408	0.8161	0.7823	0.8595	0.8564	0.7938	0.8849	0.8864	0.7920	0.8943	0.8985
c500_d100	0.6281	0.5980	0.5980	0.6734	0.6583	0.6583	0.7387	0.7186	0.7186	0.7990	0.7889	0.7889
c500_d1000	0.7989	0.7739	0.7739	0.8154	0.8249	0.8199	0.8309	0.8739	0.8704	0.8219	0.8784	0.8749
c500_d5000	0.7596	0.8489	0.8433	0.7750	0.8817	0.8819	0.7811	0.9008	0.9023	0.7823	0.9089	0.9129
c1000_d100	0.8241	0.8342	0.8342	0.8241	0.8342	0.8342	0.8241	0.8442	0.8442	0.8492	0.8593	0.8593
c1000_d1000	0.8754	0.8939	0.8934	0.8909	0.9275	0.9265	0.8849	0.9340	0.9335	0.8984	0.9455	0.9460
c1000_d5000	0.8338	0.9179	0.9195	0.8314	0.9285	0.9336	0.8397	0.9434	0.9460	0.8388	0.9472	0.9523
c1500_d100	0.9347	0.9347	0.9347	0.9548	0.9548	0.9548	0.9799	0.9749	0.9749	0.9799	0.9799	0.9799
c1500_d1000	0.9520	0.9720	0.9720	0.9575	0.9760	0.9760	0.9565	0.9770	0.9770	0.9605	0.9795	0.9795
c1500_d5000	0.9364	0.9789	0.9797	0.9431	0.9839	0.9853	0.9450	0.9867	0.9881	0.9455	0.9880	0.9889
c1800_d100	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9950	0.9950	0.9950
c1800_d1000	0.9745	0.9820	0.9820	0.9805	0.9900	0.9900	0.9810	0.9905	0.9905	0.9820	0.9910	0.9910
c1800_d5000	0.9739	0.9942	0.9945	0.9806	0.9952	0.9954	0.9802	0.9963	0.9962	0.9801	0.9963	0.9966

Table 2: Classification Accuracy on Test Set

Dataset	DecisionTree	Bagging	RandomForest	AdaBoost
c300_d100	0.6281	0.7739	0.7186	0.8241
c300_d1000	0.6723	0.8904	0.8839	0.9380
c300_d5000	0.7803	0.9185	0.9091	0.9845
c500_d100	0.6734	0.8543	0.8543	0.8693
c500_d1000	0.6818	0.8714	0.9410	0.9770
c500_d5000	0.7886	0.9432	0.9566	0.9906
c1000_d100	0.7337	0.8894	0.9749	0.9548
c1000_d1000	0.7889	0.9430	0.9910	0.9910
c1000_d5000	0.8601	0.9636	0.9949	0.9987
c1500_d100	0.8794	0.9698	1.0000	0.9899
c1500_d1000	0.9270	0.9825	1.0000	1.0000
c1500_d5000	0.9565	0.9904	0.9999	0.9999
c1800_d100	0.9146	0.9698	0.9950	1.0000
c1800_d1000	0.9780	0.9925	0.9995	0.9995
c1800_d5000	0.9873	0.9976	0.9999	1.0000

Test Set F1 Score (Weighted Average)

(technically unnecessary but useful to write the report with)

Dataset	DecisionTree	Bagging	RandomForest	AdaBoost
c300_d100	0.6263	0.7867	0.7255	0.8309

Dataset	DecisionTree	Bagging	RandomForest	AdaBoost
c300_d1000	0.7093	0.8922	0.8866	0.9386
c300_d5000	0.7884	0.9230	0.9118	0.9846
c500_d100	0.6948	0.8599	0.8543	0.8725
c500_d1000	0.6867	0.8705	0.9413	0.9770
c500_d5000	0.7981	0.9437	0.9573	0.9906
c1000_d100	0.7440	0.8866	0.9746	0.9538
c1000_d1000	0.7965	0.9436	0.9910	0.9910
c1000_d5000	0.8652	0.9635	0.9949	0.9987
c1500_d100	0.8800	0.9706	1.0000	0.9900
c1500_d1000	0.9279	0.9824	1.0000	1.0000
c1500_d5000	0.9566	0.9904	0.9999	0.9999
c1800_d100	0.9187	0.9700	0.9949	1.0000
c1800_d1000	0.9781	0.9925	0.9995	0.9995
c1800_d5000	0.9873	0.9976	0.9999	1.0000

Q&A - reflections

Decision Tree Reflection:

The selected hyperparameters balance the model complexity and generalization. For example, increasing `max_depth` or reducing `min_samples_split` and `min_samples_leaf` makes the tree more complex, which lowers bias but increases variance. The tree can fit the training data more closely, but makes it more likely to overfit. Vice versa, simpler settings reduce variance, but increase bias and generalize better for small datasets.

Fully grown trees do not always perform the best, they tend to overfit on small datasets, but help when there's sufficient data as seen with the F1 score table on Decision Trees. One pattern is accuracy, as it rose sharply from .6281 on c300_d100 to .9873 on c1800_d5000, suggesting that with more training data the

tree can learn the structure, and higher clause counts makes the classification boundary easier to separate. As clauses increase or data points increase, the decision tree has much higher F1 scores.

Bagging Reflection:

As numbers of estimators increase, validation performance generally improves, but not always with one case on `n_estimator = 50` and `100` for `max_depth None`. This shows that bagging benefits from averaging more models and reducing variance, with gains oftentimes being small and performance not necessarily increasing at each step.

Deep base trees did not always perform best. Small datasets did best with shallower tree settings, while deep trees did best on large datasets, highlighting that bagging reduces overfitting, but not always especially on small datasets.

The main pattern is that large ensembles usually help, and deep trees help when there's more data.

Random Forest:

As number of estimators increase, validation performance generally improves, but with decreasing marginal benefit after a point. More trees make the forest more stable by reducing variance, so larger ensembles generalize better.

Lowering `max_features` increases randomness and improves the ensemble as it reduces correlation between trees, but if it's too restrictive, each tree becomes weaker. Stronger restrictions increase diversity, while larger features give stronger trees.

The data showed that random forest performed extremely well, beating both single decision trees and bagging, with increases correlating with more clauses and data points.

AdaBoost Reflection

As number of estimators increases, validation performance consistently improves. Deeper base learners do not always help, if the learners become too strong, boosting loses its advantage and can overfit. Trade-off lies in weak learners that underfit and strong learners that are too aggressive.

2.5 Comparative Analysis

Which classifier achieves the best overall generalization performance? Is the ranking consistent across datasets? Explain.

Overall, AdaBoost achieved best generalization with average test accuracy at 0.9678, compared to 0.9479 for Random Forest, .9300 for Bagging, and .8167 for single Decision Tree. Ranking is consistent across most datasets, but sometimes Random Forest matched or slightly beat AdaBoost on high-performances.

- How does increasing the training data size impact performance for each classifier? Which models benefit most from more data?

Increasing training data size improved performance for all four classifiers. For example, average accuracy rose from d100 to d5000 for every method, meaning more data always helped each model learn patterns more reliably. Decision Tree showed the largest raw improvement since it had the weakest performance for small datasets.

- How does increasing the number of clauses affect performance? What does this suggest about the complexity of the underlying Boolean formula?

As number of clauses increased, performance also increased for every classifier. Average accuracy of the Decision Tree rose from about .6936 at c300 to .9600 at c1800, suggesting that formulas with more clauses were actually easier to classify. I think that adding more clauses actually makes the CNF formula more constrained, so the assignments become easier to separate.